QPCode **SET Label**

Reg.No. : .....
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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION
Semester 4
VAC- MG4VACPHY200 – ENVIRONMENTAL PHYSICS
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **25** questions

Each question carries **2** marks

1. In India, which body coordinates SDG monitoring? [K] / [CO1]

   A. Planning Commission B. NITI Aayog
   C. Ministry of Finance D. Supreme Court

2. The broadest part of an ecological pyramid represents: (CO1-U) [U] / [CO1]

   A. Apex consumers B. Herbivores
   C. Decomposers D. Producers

3. In India, mangroves are most extensive in: [K] / [CO1]

   A. Gulf of Mannar B. Sundarbans
   C. Gulf of Kutch D. Mumbai coast

4. **Assertion (A):** Excessive irrigation makes desert soil saline. [U] / [CO1]
   Reason (R): Evaporation leaves salts behind on the surface.

   A. Both A and R are true, and R is the correct explanation of A
   B. Both A and R are true, but R is not the correct explanation of A
   C. A is true, but R is false
   D. A is false, but R is true

5. Which observation made during a field visit most clearly indicates that a lake is undergoing eutrophication? [A] / [CO4]

   A. High water transparency and low nutrient concentration
   B. Abundance of submerged vegetation with clear water
   C. Frequent algal blooms accompanied by reduced water clarity
   D. Low biological productivity with minimal organic matter

6. Land becomes a non-renewable resource when: [U] / [CO1]

   A. Proper irrigation practices are followed
   B. Crop rotation is practiced regularly
   C. Dumping of toxic waste causes permanent soil contamination
   D. Organic manure is added to the soil

7. Joint Forest Management (JFM) is best described as: [U] / [CO1]

   A. Forests managed entirely by the government without local involvement
   B. Collaborative management of forests by local communities and the government
   C. Private companies managing forest resources for profit
   D. Forests left untouched without any human intervention

8. Pollution of groundwater caused by agricultural activities can be controlled by: [U] / [CO1]

   A. Use of chemical fertilizers and synthetic pesticides
   B. Using biomass as fertilizer and non-toxic pesticides such as neem
   C. Dumping untreated farm waste into water bodies
   D. Over-irrigation and improper water management

9. The Cauvery River water dispute is primarily between which of the following states? [K] / [CO1]

   A. Karnataka and Tamil Nadu
   B. Kerala and Andhra Pradesh
   C. Karnataka and Kerala
   D. Tamil Nadu and Andhra Pradesh

10. The Exxon Valdez disaster is an example of: [K] / [CO1]

    A. Accidental leakage of radioactive materials from a nuclear power plant
    B. Accidental spillage of crude oil into the ocean from a tanker
    C. Catastrophic failure of a dam causing flooding and damage
    D. Fatal accident occurring in a coal mine during mining operations

11. **Assertion (A):** Photovoltaic cells convert sunlight directly into electrical energy. [U] / [CO1]
    Reason (R): Photovoltaic cells generate electricity because sunlight heats the surface of the cell.

A. Both A and R are true, and R is the correct explanation of A
B. Both A and R are true, but R is not the correct explanation of A
C. A is true, but R is false
D. A is false, but R is true

12. Which gas has the highest global warming potential (GWP)? [K] / [CO2]

A. CO_2 B. Methane C. SF_6 D. Ozone

13. Biochemical Oxygen Demand (BOD) indicates: [K] / [CO2]

A. Mineral content B. Organic pollution level C. Salinity D. pH

14. Which pollutant persists in soil for a long time? [K] / [CO2]

A. DDT B. Compost C. Paper D. Food waste

15. At what noise level (in decibels) can hearing loss occur due to prolonged exposure? [K] / [CO2]

A. 20 B. 40 C. 70 D. 80

16. The Chernobyl nuclear disaster occurred mainly due to: [K] / [CO2]

A. Failure of safety systems and operator errors in the reactor
B. Earthquake damaging the reactor
C. Tsunami flooding the nuclear plant
D. Fire in a coal-based power plant

17. An integrated waste management strategy includes: [U] / [CO2]
A. Source reduction B. Recycling
C. Safe disposal of waste D. All of the above

18. Which of the following substances is primarily responsible for ozone layer depletion? [K] / [CO6]

A. Carbon dioxide (CO_2) B. Methane (CH_4)
C. Chlorofluorocarbons (CFCs) D. Nitrogen oxides (NO_x)

19. Acid rain is primarily caused by: [K] / [CO6]
A. Emission of sulfur dioxide and nitrogen oxides into the atmosphere
B. Release of carbon dioxide
C. Excess rainfall in coastal areas
D. Ozone layer depletion

20. Which authorities are primarily responsible for enforcing the Water (Prevention and Control of Pollution) Act in India? [K] / [CO2]

A. Pollution Control Boards
B. Ministry of Agriculture
C. Forest Department

D. Municipal Corporations only

21. The possession, trade, or transport of animal or plant parts from endangered species without proper permission is an offense under which law? [K / [CO2]

A. Wildlife Protection Act, 1972
B. Forest Conservation Act, 1980
C. Water (Prevention and Control of Pollution) Act, 1974
D. Air (Prevention and Control of Pollution) Act, 1981

22. The Montreal Protocol (1987) was signed to: [K] / [CO2]

A. Reduce CO_2 emissions
B. Prevent acid rain
C. Phase out ozone-depleting substances
D. Regulate nuclear energy

23. Eviction of tribal and other forest-dwelling communities without due process is restricted under which Act? [K] / [CO3]

A. Forest Conservation Act, 1980
B. Wildlife Protection Act, 1972
C. Forest Rights Act, 2006
D. Water (Prevention and Control of Pollution) Act, 1974

24. Which of the following is an effective strategy for managing human–wildlife interactions? [U] / [CO3]

A. Establishing wildlife corridors and buffer zones
B. Encouraging expansion of agriculture into forest areas
C. Eliminating wild animals from conflict regions
D. Ignoring conflicts as natural events

25. Which factor best explains the link between population growth and climate change? [U] / [CO3]
A. Increased heat from the Sun
B. Higher energy consumption and greenhouse gas emissions
C. Change in Earth's magnetic field
D. Decrease in atmospheric pressure

26. Resettlement of people displaced by large dams becomes challenging mainly due to: [U] / [CO3]
A. Lack of land only
B. Loss of livelihood, cultural ties, and established social networksC. Excessive rainfall
D. High electricity production

27. Landslides are more frequent after heavy rainfall because: [U] / [CO5]
A. Rain increases soil cohesion
B. Water reduces friction between soil particles
C. Rocks become heavier permanently

D. Gravity decreases

28. The Silent Valley project was stopped mainly due to concerns about: [U] / [CO5]
    A. Electricity shortage
    B. Loss of unique biodiversity
    C. Flood control failure
    D. High construction cost

29. Environmental ethics refers to: [U] / [CO6]
    A. Rules for industrial growth
    B. Moral principles guiding human–nature relationship
    C. Economic laws of resource use
    D. Political control of environment

30. Public awareness plays a key role in environmental protection because it: [U] / [CO6]
    A. Eliminates natural disasters
    B. Encourages informed and responsible behaviour
    C. Replaces scientific research
    D. Stops population growth immediately

[2x25 = 50]